

CT Scanner Hardware Subsystems & Troubleshooting Data

A engineering and technical reference document mapping core hardware components, functional descriptions, common failure modes, diagnostic codes, and textbook references.

1. Hardware Subsystem & Error Architecture Data

Subsystem	Component	Common Failure / Symptom	Error Code	Textbook Reference Module
Gantry	X-Ray Tube	Thermal overload, anode arcing, filament degradation	ERR-XRAY-401	X-Ray Production & Tube Heating Storage Capacity
Gantry	Slip Rings	Brush wear, data transmission errors, optical link dropouts	ERR-SLIP-102	Gantry Design & Continuous Rotation Systems
Gantry	HV Generator	KV switching faults, high-voltage inverter failures, oil leaks	ERR-HVG-305	Power Subsystems & High-Frequency Generators
Data Chain	Detectors & DAS	Channel calibration drift, ADC errors, temperature fluctuations	ERR-DAS-208	Data Acquisition Systems & Multi-Slice Detectors
Patient Handling	Couch/Table	Horizontal drive motor stall, positioning encoder mismatch	ERR-TBL-501	Patient Positioning Mechanics & Safety Controls
Computing	Recon Engine	Parallel processing node timeout, backprojection artifact errors	ERR-REC-099	Image Reconstruction & Array Processor Hardware

2. Core Preventive Maintenance Intervals

Subsystem	Maintenance Task	Frequency	Impact of Neglect
Slip Rings	Clean carbon brush debris and inspect optical fiber links	Monthly	Data packet loss, image artifacts, or total reconstruction aborts
X-Ray Tube	Perform tube calibration, air calibrations, and rotor checks	Weekly	Baseline calibration drift, phantom ring artifacts
Cooling System	Inspect heat exchanger, fluid levels, and fans	Quarterly	Tube overheating, rapid thermal cutout, shortened tube lifespan
Gantry Mechanical	Check drive belts, safety interlocks, and emergency stops	Semi-Annually	Mechanical slippage, physical safety hazards, geometric misalignment

Disclaimer: This document serves as a standard data reference blueprint for engineers studying biomedical imaging equipment architecture. Field operations must strictly align with manufacturer manuals (e.g., GE, Siemens, Philips, Canon) and specific site protocols. This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.